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# Technical Specification for Quote Generation Web Application

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# Objective

Develop a web-based application that allows users to input transaction details from client invoices and calculate potential cost savings if they switch to Transact World. The application will generate a PDF quote with a breakdown of transaction details, cost comparison, and savings.

## Core Functionalities

### User Inputs

#### Transaction Data Input

- Upload or manually input invoice data (CSV/Excel or form fields).
- Fields:
  - Total transaction volume.
  - Number of transactions per card type.
  - Total volume processed per card type.
  - Date range (usually monthly).

#### Business Type Selection

- Dropdown to select business type (e.g., Retail, E-commerce, Hospitality).
- Each business type will have predefined markup rates.

#### Volume and Transaction Details

- Input total transaction volume and number of transactions.
- Adjust markup rates based on volume:
  - Higher volume = lower markup.
  - Lower volume = higher markup.

#### Markup Rate Configuration

- Allow manual entry of markup rates for each card type.
- Markup rates will have two components:
  - Percentage markup on calculated cost.
  - Fixed markup on authorization fee (£0.05 per transaction).

#### Card Type and Issuing Location

- Dropdown to select card type (e.g., Visa Credit, Mastercard Debit).
- Dropdown to select card issuing location (e.g., UK & EEA, International)

# Calculations

## Cost Calculation

For each card type and issuing location:

- **Interchange Fee:** Based on card type and issuing location (e.g., 0.30% for Visa Credit UK & EEA).
- **Scheme Fee:**
  - Percentage fee (e.g., 0.0230% for Visa Credit).
  - Fixed fee (e.g., £0.0097 for Visa Credit).
- **Acquirer Bank Fee:** 0.08% of transaction value.
- **Authorization Fee:** £0.05 per transaction.

## Total Cost Calculation

Sum of:

- Interchange fee.
- Scheme fee (percentage + fixed).
- Acquirer bank fee.
- Authorization fee.

## Markup Calculation

Apply markup rates:

- Percentage markup on total cost.
- Fixed markup on authorization fee.

## Savings Calculation

- Compare client's current costs with TransactWorld's proposed costs.
- Calculate monthly and annual savings.

## Output

- Generate a detailed PDF report showing:
  - Information received from the client.
  - Breakdown of card transaction details.
  - Comparative analysis of costs.
  - Visual representation (table + infographic) of savings.

## Data Model

### Card Types:

Fields: Card type (e.g., Visa Credit), issuing location, interchange fee, scheme percentage fee, scheme fixed fee.

### Business Types:

Fields: Business type (e.g., Retail), markup rates.

### Transaction Data:

Fields: Total volume, number of transactions, card type, issuing location, date range.

## Admin Panel

- Manage acquirers and authorised transaction rate
- Manage Payment method (card type)
- Manage interchange fees & scheme fees
- Set default markup rates per business type
- Add customer and generate quotes
- View history of generated quotes
- Export Quotes

## Example Calculation

### Input:

- Card Type: Visa Credit , Issuing Location: UK & EEA.
- Transaction Volume: £10,000 and Number of Transactions: 100.

### Calculations:

- Interchange Fee:  $0.30\%$  of £10,000 = £30.
- Scheme Fee:  $0.0230\%$  of £10,000 +  $£0.0097 \times 100$  = £2.30 + £0.97 = £3.27.
- Acquirer Bank Fee:  $0.08\%$  of £10,000 = £8.
- Authorization Fee:  $£0.05 \times 100$  = £5.
- Total Cost: £30 + £3.27 + £8 + £5 = £46.27.
- Markup:  $1.5\%$  of £46.27 +  $£0.02 \times 100$  = £0.69 + £2 = £2.69.
- Total Proposed Cost: £46.27 + £2.69 = £48.96.

### Savings:

- If client's current cost is £60, savings = £60 - £48.96 = £11.04 per month.

## Technology Stack

- **Frontend:** React.js / Vue.js or PHP
- **Backend:** Node.js / Python (Django/FastAPI)
- **Database:** MySQL
- **File Processing:** Pandas (for CSV/Excel parsing)
- **PDF Generation:** ReportLab / jsPDF

## Next Steps for Development

1. **Define UI Wireframes**
  - Layout for input form, quote display, admin panel
2. **Set Up Database Schema**
  - Tables for transaction details, markup rates, generated quotes
3. **Develop Cost Calculation Engine**
  - Implement formulas to compute costs dynamically
4. **Integrate PDF Generation**
  - Custom template with branding & infographic visualization
5. **Testing & Deployment**
  - Validate calculations with real client data
  - Optimize UX for smooth data entry & adjustments

## Expected Deliverable

- Fully functional web application with an intuitive UI
- Automated cost calculation & quote generation
- Secure admin panel for managing rates & quotes
- Hosted application with database & PDF export functionality

## Example of Web / Quote online from PaymentSense

Pricing This is an excellent example of what I want to have on our site, how ever our calculations are needed to be more informative.

Link from : <https://www.paymentsense.com/uk/pricing/>

### Pricing

Fill out the details below, and we'll find the best deal for your business.

|                                                  |         |
|--------------------------------------------------|---------|
| Your total card acceptance in previous 12 months | £352000 |
| Your average transaction value                   | £50     |
| Cafe, Brasserie & Restaurant                     | ▼       |

Show pricing

If your annual card over is above £150K, our team can advise on bespoke pricing

Get a customised quote

#### Your costs for accepting common individual card payments<sup>5</sup>

| MasterCard             |                    |                     |                       | Visa                   |                    |                     |                       | Additional charges per transaction |    |
|------------------------|--------------------|---------------------|-----------------------|------------------------|--------------------|---------------------|-----------------------|------------------------------------|----|
| UK-based Card          | Debit <sup>2</sup> | Credit <sup>3</sup> | Business <sup>4</sup> | UK-based Card          | Debit <sup>2</sup> | Credit <sup>3</sup> | Business <sup>4</sup> | Authorisation fee                  | 5p |
| In person <sup>6</sup> | 1.40%              | 1.40%               | 1.99%                 | In person <sup>6</sup> | 1.40%              | 1.40%               | 1.99%                 |                                    |    |
| Online                 | 1.40%              | 1.40%               | 1.99%                 | Online                 | 1.40%              | 1.40%               | 1.99%                 |                                    |    |
| Phone <sup>7</sup>     | 1.40% + 50p        | 1.90%               | 2.49%                 | Phone <sup>7</sup>     | 1.40% + 50p        | 1.90%               | 2.49%                 |                                    |    |

|                                     |                |        |            |     |
|-------------------------------------|----------------|--------|------------|-----|
| Other potential transaction charges | Chargeback fee | £28.00 | Refund fee | 50p |
|-------------------------------------|----------------|--------|------------|-----|

#### Example

|                                                                                                                |                                |                                  |             |
|----------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------|-------------|
| If your customer uses a Mastercard <b>personal debit card in person</b> to purchase a £10 item, you would pay: | <b>14p</b><br>(1.4% of £10.00) | <b>5p</b><br>(authorisation fee) | <b>=19p</b> |
| If your customer uses a Visa <b>personal credit card by phone</b> to purchase a £10 item, you would pay:       | <b>19p</b><br>(1.9% of £10.00) | <b>5p</b><br>(authorisation fee) | <b>=24p</b> |

#### Other monthly charges

|                                                     |               |
|-----------------------------------------------------|---------------|
| Point of sale terminal (per terminal per month)     | £20.00 (+VAT) |
| Gateway <sup>8</sup>                                | £9.95         |
| PCI DSS service fee <sup>9</sup>                    | £4.95 (+VAT)  |
| Minimum monthly service charge (MMSC) <sup>10</sup> | £24.95        |
| Account on File Fee (per month) <sup>11</sup>       | £3.99         |
| PCI Non-Compliance (per month)                      | £35.00 (+VAT) |
| Reporting Fee (per month) - free for 3 months       | £12.95        |

#### Other details

|                         |                                        |
|-------------------------|----------------------------------------|
| Settlement time         | 24 working hours                       |
| Payment brands accepted | VISA<br>MasterCard<br>American Express |

# Additional Information

## Example of Customer Invoice

Merchant ID 202020315720221  
Invoice number GB01386031  
Invoice period 07 Jul to 06 Aug 2024  
Invoice date 07 Aug 2024

### Card transactions

| Fee type                           | Quantity | Fee per transaction       | Total   | VAT code |
|------------------------------------|----------|---------------------------|---------|----------|
| Card transaction charges           | -        | See breakdown table below | £743.93 | E        |
| Secure Transaction Fee             | 2191     | £0.04                     | £87.64  | S        |
| Refund Fee                         | 7        | £0.50                     | £3.50   | E        |
| Total transaction fees and charges |          |                           | £835.07 |          |



## Example of Breakdown of transactions

Invoice period                      07 Jul to 06 Aug 2024  
 Invoice date                        07 Aug 2024

### Card transaction rates breakdown

| Card type                                          | Number of transactions | Total value of transactions | Rate per transaction | Total          | VAT code |
|----------------------------------------------------|------------------------|-----------------------------|----------------------|----------------|----------|
| American Express                                   | 242                    | £10,513.10                  | 1.9%                 | £199.75        | E        |
| Mastercard Credit                                  | 360                    | £13,853.15                  | 0.99%                | £137.15        | E        |
| Visa Debit (inc. prepaid)                          | 836                    | £20,910.63                  | 0.59%                | £123.37        | E        |
| Mastercard Debit (inc. prepaid)                    | 628                    | £13,033.40                  | 0.59%                | £76.90         | E        |
| Visa Credit                                        | 189                    | £7,611.35                   | 0.99%                | £75.35         | E        |
| Mastercard Corporate and Purchasing                | 19                     | £2,860.40                   | 1.99%                | £56.92         | E        |
| Visa Business Debit                                | 28                     | £1,824.65                   | 1.99%                | £36.31         | E        |
| Mastercard Business                                | 39                     | £931.46                     | 1.99%                | £18.54         | E        |
| Visa Corporate and Purchasing                      | 5                      | £309.10                     | 1.99%                | £6.15          | E        |
| Visa Credit, International                         | 2                      | £125.40                     | 1.94% + £0.13        | £2.69          | E        |
| Visa Debit (inc. prepaid), Non-qualifying          | 2                      | £200.00                     | 1.09%                | £2.18          | E        |
| Visa Corporate and Purchasing, International       | 5                      | £49.80                      | 2.94% + £0.13        | £2.11          | E        |
| Visa Credit, Non-qualifying                        | 1                      | £100.00                     | 1.49%                | £1.49          | E        |
| Mastercard Corporate and Purchasing, International | 1                      | £44.60                      | 2.94% + £0.13        | £1.44          | E        |
| Mastercard Credit, International                   | 3                      | £44.90                      | 1.94% + £0.13        | £1.26          | E        |
| Visa Business                                      | 1                      | £50.00                      | 1.99%                | £1.00          | E        |
| Mastercard Debit (inc. prepaid), International     | 3                      | £15.50                      | 1.54% + £0.13        | £0.63          | E        |
| Visa Debit (inc. prepaid), International           | 1                      | £16.95                      | 1.54% + £0.13        | £0.39          | E        |
| Visa Debit (inc. prepaid), EEA                     | 4                      | £46.55                      | 0.64%                | £0.30          | E        |
| <b>Subtotal</b>                                    | <b>2369</b>            | <b>£72,540.94</b>           |                      | <b>£743.93</b> |          |
| <b>Total transaction charges</b>                   |                        |                             |                      | <b>£743.93</b> |          |

## Interchange rates ( Page 1 and 2) also the additionl fee in Page 2

### Acquiring Interchange and Scheme Fee guide

| Credit category           | Card type                       | Card issued   | Interchange fees             | Scheme fees       |
|---------------------------|---------------------------------|---------------|------------------------------|-------------------|
| Credit cards              | Visa Credit                     | UK & EEA      | 0.30%                        | 0.0230% + £0.0097 |
|                           |                                 | International | 0.30%-1.50%                  | 0.0230% + £0.0930 |
|                           | Mastercard Credit               | UK & EEA      | 0.30%                        | 0.0298% + £0.0085 |
|                           |                                 | International | 0.30%-1.50%                  | 0.0298% + £0.0914 |
| Debit cards               | Visa Debit (inc. Prepaid)       | UK & EEA      | 0.20%                        | 0.0180% + £0.0097 |
|                           |                                 | International | 0.20%-1.15%                  | 0.0180% + £0.0930 |
|                           | Visa Electron                   | UK & EEA      | 0.20%                        | 0.0180% + £0.0097 |
|                           |                                 | International | 0.20%-1.15%                  | 0.0180% + £0.0930 |
|                           | Visa V Pay                      | UK & EEA      | 0.20%                        | 0.0180% + £0.0097 |
|                           | Mastercard Debit (inc. Prepaid) | UK & EEA      | 0.20%                        | 0.0298% + £0.0085 |
|                           |                                 | International | 0.20%-1.15%                  | 0.0298% + £0.0914 |
|                           | Maestro (inc. Prepaid)          | UK & EEA      | 0.20%                        | 0.0298% + £0.0085 |
|                           |                                 | Europe        | 0.20%-1.15%                  | 0.0298% + £0.0085 |
|                           |                                 | International | 0.60%-1.60% +<br>€0.00-€0.25 | 0.0298% + £0.0914 |
| Visa Business Debit cards | Visa Business Debit             | UK            | 0.50%-1.40%                  | 0.0180% + £0.0097 |
|                           |                                 | EEA           | 1.30%-1.90%                  | 0.0180% + £0.0097 |
|                           |                                 | Europe        | 1.60%-2.00%                  | 0.0180% + £0.0097 |
| Other business cards      | Visa Corporate and Purchasing   | UK            | 1.70%-2.00%                  | 0.0230% + £0.0097 |
|                           |                                 | Europe        | 1.80%-2.00%                  | 0.0230% + £0.0097 |
|                           |                                 | International | 2.00%                        | 0.0230% + £0.0930 |
|                           | Visa Business                   | UK            | 1.40%                        | 0.0230% + £0.0097 |
|                           |                                 | Europe        | 1.30%-2.00%                  | 0.0230% + £0.0097 |
|                           |                                 | International | 2.00%                        | 0.0230% + £0.0930 |
|                           | Mastercard Business             | UK            | 0.70%-1.70%                  | 0.0298% + £0.0085 |
|                           |                                 | EEA           | 0.80%-1.65%                  | 0.0298% + £0.0085 |
|                           |                                 | Europe        | 0.80%-2.10%                  | 0.0298% + £0.0085 |
|                           |                                 | International | 2.00%                        | 0.0298% + £0.0914 |
|                           | Maestro Business and Commercial | UK            | 0.40%-1.65% +<br>£0.00-£0.05 | 0.0298% + £0.0085 |
|                           |                                 | EEA           | 0.40%-1.65% +<br>£0.00-£0.05 | 0.0298% + £0.0085 |

|                                             |                                |               |             |                   |
|---------------------------------------------|--------------------------------|---------------|-------------|-------------------|
|                                             |                                | Europe        | 0.80%-2.10% | 0.0298% + £0.0085 |
| Diners Club<br>International (DCI)<br>cards | DCI Credit                     | UK & EEA      | 0.30%       | 0.1500%           |
|                                             |                                | International | 1.50%       | 0.1500%           |
|                                             | DCI Debit (inc. Prepaid)       | UK & EEA      | 0.20%       | 0.1500%           |
|                                             |                                | International | 1.50%       | 0.1500%           |
|                                             | DCI Business and<br>Commercial | UK & EEA      | 1.75%       | 0.1500%           |
|                                             |                                | International | 1.75%       | 0.1500%           |

## Additional scheme fees

|                                                                             |        |
|-----------------------------------------------------------------------------|--------|
| Mastercard/Maestro Europe Cross Border Transactions                         | 0.045% |
| Mastercard/Maestro International Cross Border Transactions                  | 0.45%  |
| Mastercard/Maestro Europe Card Not Present Cross Border Transactions        | 0.05%  |
| Mastercard/Maestro International Card Not Present Cross Border Transactions | 0.45%  |
| Visa Card Not Present Transactions                                          | 0.01%  |
| Visa International Cross Border Transactions                                | 0.45%  |
| Visa International Card Not Present Cross Border Transactions               | 0.40%  |
| Mastercard Ecomm 3D Secure Requests                                         | €0.044 |
| Visa Ecomm 3D Secure Requests                                               | €0.020 |

## Definitions

|                      |                                                                        |
|----------------------|------------------------------------------------------------------------|
| <b>UK</b>            | Card issued in the UK                                                  |
| <b>EEA</b>           | Card issued outside the UK but in the European Economic Area (EEA)     |
| <b>Europe</b>        | Card issued outside the EEA but within the Card Schemes' Europe region |
| <b>International</b> | Card issued outside the Card Schemes' Europe region                    |